

Coast Happiness Project — Technical Report

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1. Problem and scope

Hong Kong has roughly 850km of coastline and persistent marine debris loading, but the public-facing tooling for "where is trash likely to accumulate today, and should I clean there?" is essentially nil. Predictive water-quality work has plenty of precedent in your group (beach-side FIB nowcasts, hydrodynamic-driven forecasts) — there is much less for marine debris, where the physical signal is dominated by surface wind and surface currents rather than runoff microbiology.

The system is a low-cost, observational-data-only predictor for relative trash accumulation across 86 beaches and 6 documented hotspots in Hong Kong, packaged as a public-facing app that also schedules and de-duplicates community cleanups.

2. Model

The per-beach trash index $S \in [0, 1]$ is computed every 6 hours as:

$$S = \text{clamp}(0, 1, (\text{dynamic} + 0.25 \cdot H) \cdot M_{\text{bay}})$$
$$\text{dynamic} = (0.45 \cdot W + 0.15 \cdot V + 0.15 \cdot R) \cdot E$$
$$M_{\text{bay}} = 1 + 0.3 \cdot b$$

with components:

Symbol	Meaning	Source
W	Onshore wind score	HKO 10-min wind, station-blended
V	Onshore wave score · height factor	HKO marine bulletin
R	River-mouth proximity (exponential decay, $\tau=8\text{km}$)	Static catalogue of 18 river mouths
H	Historical accumulation prior (0–1)	Hand-tuned from cleanup reports
E	Exposure to open sea (0–1)	Hand-tuned per beach
b	Bay trap-factor (0–1)	Hand-tuned per beach

Coefficients are not learned — they reflect physical intuition (wind dominates surface debris transport, historical priors anchor against unmeasured site-specific quirks). They are the obvious thing to optimize against ground-truth observations once we have them; see §6.

2.1 Onshore wind score

For a station with wind from direction θ_w and a beach facing the sea at orientation θ_b , the onshore component is:

$$\text{onshore}(\theta_w, \theta_b) = \max(0, \cos(|\theta_w - \theta_b|))$$

Each beach is associated with its 2 nearest HKO stations (mapping hand-curated for the 39 "original" beaches and inherited via 3-nearest-neighbour distance-weighted assignment for the 47 added from the hkcleanup.org coastal map). The wind score blends per-station onshore $\times$ speed $\times$ max-speed-normalisation across that pair, plus a gust bonus when $\text{gust} - \text{mean} > 0$.

2.2 Wind smoothing

Live wind alone overreacts to lulls. Predictions use an exponentially decay-weighted integration of the last 72h of snapshots with an 18-hour half-life:

$$w(\Delta t) = \exp(-\Delta t / \tau), \quad \tau = 18\text{h} / \ln 2$$

Direction is averaged as vectors:

$$\begin{aligned} \bar{u} &= \sum_i w_i \cdot s_i \cdot \sin(\theta_i), & \bar{v} &= \sum_i w_i \cdot s_i \cdot \cos(\theta_i) \\ \bar{\theta} &= \text{atan2}(\bar{u}, \bar{v}), & \bar{s} &= \sqrt{\bar{u}^2 + \bar{v}^2} \end{aligned}$$

so $350^\circ + 10^\circ$ collapses to North rather than 180° . Snapshots are written to Firestore on a 6-hour Cloud Function schedule (`collectWind`).

A "recent storm boost" flag is raised when the smoothed onshore score exceeds $1.3 \times$ the current instantaneous score — this surfaces beaches where a passing front already pushed debris in, even if conditions have since calmed.

2.3 Wave and river components

Wave: HKO marine bulletin gives wave height and direction. Score is $\text{onshore}(\theta_{\text{wave}}, \theta_b) \cdot \min(1, h/3)$. Capping at 3m prevents typhoon heights from dominating.

River: simple exponential-decay sum of distance-weighted weights from 18 documented river mouths (Pearl River outflow, Shing Mun, Lam Tsuen, Tin Shui Wai, etc.) with decay constant $\tau = 8$ km. The river term is exposure-gated like the wind/wave terms.

2.4 Bay multiplier

Empirically motivated: debris that drifts into a high- b bay (Cheung Sha Lan, Hap Mun Bay) leaves much more slowly than from a straight shoreline. We model this as a simple $(1 + 0.3 \cdot b)$

multiplier on the final score, applied *after* the exposure-gated dynamics. `b` is a hand-tuned 0–1 indicator.

3. Data

Source	Coverage	Cadence
HKO <code>latest_10min_wind.csv</code>	30 stations across Hong Kong	10 min
HKO marine bulletin (waveHeight, waveDirection, wavePeriod)	Aggregated near-Hong Kong	Hourly
Beach catalogue	92 entries, bilingual EN/zh-HK	Static
hkcleanup.org Coastal Map markers	94 markers (1 outlier rejected)	Static, refresh on demand
HandsOn HK volunteer opportunities	"Environmental Conservation" filter	Scraped per request, 1h cache

Beach features are stored as JSON. Each row has `name`, `nameTc`, `lat`, `lng`, `region`, `orientation` (deg), `exposure`, `bayFactor`, `historicalWeight`, `nearestStations` (array of HKO names), `accessDifficulty` (1–5), `roadTier` $\in$ {road, ferry, kaito, remote}, `hikeGrade` $\in$ {flat, rolling, steep, scramble}, and a parsed `debris` object (cleaning frequency, transport notes, age suitability, coastal length — sourced from hkcleanup's site descriptions).

4. Beach catalogue construction

Two principled passes:

1. **Anchor on hkcleanup.org Coastal Map** — 93 surviving markers used as the canonical site list with their coordinates. Matched against an existing hand-tuned catalogue by Chinese name (most reliable), then English name. Proximity matching was deliberately excluded after it caused silent identity merges between adjacent beaches (Kwun Yam $\leftrightarrow$ Cheung Chau Tung Wan, both at ~250m apart).
2. **Add 7 hotspots not on the canonical map but documented in cleanup literature** — Po Chu Tam, Sandy Bay, Cheung Sha Lan (corrected coordinates after spotting a Discovery Bay vs South Lantau mix-up in the source data), Sam Pak Wan, Lo Tik Wan, Mui Wo, Chek Keng, Peng Chau Tung Wan.

For matched beaches, engine parameters are inherited verbatim. For the 47 new beaches, engine parameters are derived from the 3 nearest tuned neighbours, distance-weighted:

- `orientation` : unit-vector average of neighbour orientations.
- `exposure` , `bayFactor` : weighted scalar average.

- `nearestStations` : top-2 most-cited neighbour stations.
- `accessDifficulty` , `roadTier` , `hikeGrade` : parsed from the `hkcleanup` description blob when present (Low/Middle/High maps to 1/3/4, with bumps for "remote" transport).

This is best-effort and explicitly flagged as a limitation — a beach that genuinely faces an opposite aspect than its neighbours will be mispredicted until tuned by hand.

5. System and engineering

- **Stack:** Node.js (Express) backend, React + Leaflet frontend, Firebase Hosting + Cloud Run (us-central1), Firestore for wind history. Scheduled Cloud Function pulls wind snapshots every 6h.
 - **Lines of code:** ~3k server, ~3k client, ~700 in build scripts.
 - **Cost:** Currently within Firebase free tier. Cloud Run scales to zero between requests; cold-start latency on the first hit of the day is ~3s.
 - **Bilingual:** English / Traditional Chinese throughout. Map tile layer is language-aware (CARTO Voyager for EN labels, standard OSM for the Chinese-first HK `name` tags).
 - **Other features:** integration with HandsOn HK's environment category as a public cleanup feed (Plastic Free Seas, Green Power, Soap Cycling); cleanup hosting flow with auto-deduplication against upcoming HandsOn events at the same beach; recycling drop-off map for the 130 GREEN@COMMUNITY stations.
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6. Limitations and the questions I'd most value your input on

Validation

The most glaring gap is the absence of any quantitative validation. We've sanity-checked outputs against known hotspots (Cheung Sha Lan, Pak Nai, Tai Long Sai Wan stay flagged in onshore-wind regimes), but there's no formal evaluation against measured debris loads.

Plastic Free Seas have site-specific cleanup quantification — kilograms collected, item types — for several PFS-managed sites. The natural next step is to align that record with daily predicted indices over the same period and compute (a) Pearson / Spearman correlation, (b) AUC against a discretised "high-debris-day" threshold, (c) lead-time analysis (does predicted "high" yesterday predict observed "high" today?).

Question 1: Given the small N and irregular temporal spacing of cleanup events, what would you use for a defensible validation framework? Spatial cross-validation (leave-one-beach-out)? Bootstrap on event-day records?

Coefficient identifiability

Currently the 0.45/0.15/0.15/0.25 split is hand-set. Three identifiability concerns:

- Wind and wave directions are highly collinear in HK (both forced by the prevailing monsoon).
- The historical prior H is itself derived from documented debris reports, which probably correlate with the dynamic terms via unobserved confounders (e.g. all the same beaches are wind-exposed *and* historically dirty).
- Exposure E gates the dynamic component multiplicatively — if you set $E=0$ everything but H zeroes out.

Question 2: Given a small validation set, would a Bayesian hierarchical model (partial pooling across beaches within region) be overfit relative to the data, or is it the right move to identify the wind/wave contributions separately? Any priors you'd recommend?

Surface drift physics

The cosine-of-direction-difference is a crude proxy for surface transport. A more principled approach would couple to a HF radar or HYCOM-derived surface current product near HK waters. The fetch distance also matters more than direction at low wind speeds.

Question 3: Is there a "good enough" public-domain surface current product for the South China Sea margin that would be practical to integrate without becoming a research project in itself? Or is the lift not worth it for a marginal accuracy gain?

Spatial coverage of wind stations

The 30 HKO stations cluster near Victoria Harbour and the airport. Remote shorelines (Tai Long Sai Wan, Po Chu Tam) are paired with the closest 1–2 stations, but those stations are 8–15 km away with intervening topography. Idealised numerical mesoscale wind (WRF, ~1 km resolution over HK) exists but the licensing path through HKO is unclear.

Question 4: When the closest measurement is meaningfully mismatched in topographic regime, is it more honest to (a) refuse to score the beach, (b) widen the uncertainty band on its score, or (c) fall through to historical priors only?

7. What I'd hope to get from a conversation

The system is in production but the predictive science is the weakest part. I'd value (in rough priority order):

1. Methodological guidance on validating without a clean labelled dataset.
2. Pointers to similar work in your group or community — predictive beach-condition models that have been deployed to the public, and what worked / didn't in terms of communicating uncertainty to lay users.
3. Honest scepticism on whether the model architecture is even the right shape for the problem, vs. a different formulation (e.g. Lagrangian particle tracking starting from river mouths and monsoon trade winds).
4. If there's a Stanford research thread this could plausibly connect to as undergrad/master's-level work — anything in your group on marine debris prediction or

even non-debris coastal forecasting where the methodology transfers.

The code, data, and historic predictions are open and available on request. Happy to share a runbook for reproducing the model offline.